

EARTH 201: Introduction to Physical Geography

Updated August 06, 2026

(1528 North University Building -NUB)

Instructor: Prof. Jessica V. Fayne jfayne@umich.edu (office hours Wednesday 10-11:00, Zoom; In person 30 minutes before class, 1528 NUB)

Lecture Hours: 11:30-12:50 pm Tuesday & Thursday

Canvas class web site: CANVAS

Overview: This introduction to physical geography emphasizes the nature and dynamics of the Earth system including the atmosphere, hydrosphere and solid Earth, along with their interactions. Topics include weather systems, climate change, biogeography, soils, plate tectonics, erosion, freshwater resources, and landforms, all of which are discussed in the context of Earth Systems Science.

This course provides an introduction to Earth and Environmental Sciences for science majors, but is also a science course that is relevant to social science and humanities majors who wish to be well-informed global citizens, or who simply want to better understand their natural surroundings.

Physical Geography is a highly integrative discipline that seeks to understand the many natural processes taking place at or near Earth's surface. The intersection of these processes provides a physical setting for human activities. Sometimes referred to as Earth System Science, Physical Geography is therefore essential to fully understand human geography, culture, politics and economics.

It is our expectation that everyone successfully completing this course will master the scientific principles and vocabulary for understanding processes taking place in the surface environment and be able to use this knowledge to understand current events related to climate change, weather, and tectonics. We will accomplish this goal with lectures, selected readings, and directed discussion with laboratory exercises.

Learning objectives: With Bloom's Taxonomy

Bloom's Taxonomy is a simplified method for describing and categorizing educational goals and learning skills. The Taxonomy includes six tiers, from fundamental and introductory skills to advanced mastery. The six tiers include 1) **Remembering** facts, terms and basic concepts, 2) **Understanding** by being able to paraphrase and re-explain concepts, 3) **Applying** learned skills when prompted, given similar conditions 4) **Analyzing** concepts to understand patterns, relationships, and connections, 5) **Evaluating** the practicality or utility of the concept, given relevant (or irrelevant) situations, 6) **Creating** new ideas by synthesizing distinct and relevant concepts

As this is an introductory course, we will focus on the first three concepts 1) Remember, 2) Understand, 3) Apply.

To meet these objectives, the course is organized as follows:

- 1) Remember:
 - a. Weekly In-lecture short vocab quizzes will count for participation/attendance grade
 - b. Virtual "field-trips" will also count for participation/attendance
- 2) Understand:
 - a. 11 Weekly lab assignments
- 3) Apply:
 - a. Three exams on vocabulary with short answer prompts

Required Textbooks

- Textbook: Nesar, Laura. January 30, 2025. Introduction to Earth Science. 2nd Ed. (ISBN 978-1-962841-24-5)

- Workbook (lab assignments): Hess, Darrel. May 13, 2021. McKnight's Physical Geography: A Landscape Appreciation --Physical Geography Laboratory Manual. 13th Ed. (ISBN 9780135918128)

~ Textbook links

- The Main Textbook is FREE: <https://vtechworks.lib.vt.edu/items/ea151149-ff36-420a-891a-99ea8aa79d27>
- The Main Workbook is not free: <https://www.pearson.com/en-us/subject-catalog/p/physical-geography-laboratory-manual/P200000006961/9780135923900> copies range in price from **online rental**: \$65/semester to **print**: \$90-115
- GSI Reference Textbook (is not required for students) is not free: <https://www.pearson.com/en-us/subject-catalog/p/mcknights-physical-geography-a-landscape-appreciation/P200000006987> copies range in price from **online rental**: \$65/semester to **print**: \$90-115

How much time will you spend on labs?

Labs will be started in the classroom during the lab section and *may be* completed outside of the classroom. There are some labs that strongly link with other courses available in our department – these labs might have more content (longer) to enable students to more comfortably transition into those courses. However, there are some other topics (such as atmospheric science) that are included in “physical geography”, but not “earth science”, and therefore, we will spend a little less time on those topics. Most of the labs can be completed within the classroom, however some labs might require the use of a computer, which students may complete using their personal computers or using university public computers (such as those in Angell Hall). **Students should expect to spend up to 3 hours per week outside of the classroom on labs and studying for exams.**

The majority the lab content comes from the “McKnight's Physical Geography: A Landscape Appreciation Lab Manual”. Each lab consists of 3-6 ‘Exercises’; each of the ‘Exercises’ has 2-10 questions and takes about 5-10 minutes to complete. **‘Exercises’ may be supplemented by additional material using real-world data and Google Maps.**

How will exams work?

There are three exams in the course. **All exams are online, on canvas, and students have 24 hours to complete the exams. Each exam consists of primarily multiple choice and a few short answer responses.** The first two exams are expected to take 30-60 minutes, and the final exam is expected to take 60-120 minutes. Exams are online and students may complete their exams anywhere (even at home), but **exams are closed-book**. Students who require additional accommodation are expected to submit their accommodations request to SSD at the beginning of the semester (read below for details on SSD accommodations).

30% of the course grade is from attendance?!? Is that right?

That's right! We want to encourage regular attendance in both lecture and lab to build good academic habits and to help reinforce learning. Much of the learning comes from hearing, understanding, and using terminology in-context, so we believe that attending classes regularly will support learning and will make completing the assignments much easier and more intuitive. Short quizzes will be available throughout the semester as completion-participation grade (you get credit for doing it!). These short quizzes are designed to reinforce topics discussed in previous lectures and lab and to prepare you for the upcoming exams.

What should you bring to class?

- Electronic device for following along with the lecture notes and online textbook:
 - Cell phone, Laptop, Tablet, etc.
- Physical notebooks (and pen/pencil) are optional, but you might find them useful for drawing diagrams
- A positive attitude!

Classroom Etiquette

Students are expected not to talk while the instructor is lecturing. During work times, however, students may talk to each other or use earbuds. Students may use personal phones and laptops in class, as long as these devices are not distracting to the rest of the class. Distractions would include playing audio or looking at unrelated images, or disrupting classmates. Students who behave in a disruptive manner will be asked to leave the room for the remainder of the class period.

All students, GSI, and the Professor are important and equal partners in this education process regardless of racial/ethnic, cultural, gender, age, social class, caste and other kinds of human difference. We highlight the importance of inclusivity and respect for all students and students' contributions.

Equal partnership requires mutual goodwill and positive intent. Just as Instructors must come to class prepared to answer questions and provide guidance, students must complete assignments on time (or early in case there are any questions). Questions regarding the assignments are welcome, provided that students demonstrate they have completed the pre-requisite reading which explains the assignment.

Attendance Policy

Attendance at the lecture and lab is expected and worth 30% of the final grade. However, there are exceptions for religious holidays, and emergencies.

- If you have a religious holiday, please communicate with the instructor and the GSI (and group members, if applicable) at least 1 week in advance to make plans to stay on track in the course.
- If you are experiencing an emergency, please communicate with the instructor and GSI as soon as possible (after the emergency has resolved, and to your ability) to make plans to stay on track in the course.
- If you experience a major conflict or life emergency that requires more than two class periods of absence, you will need to provide documentation, such as a doctor's note, and meet with the instructor either in person or over Zoom to create a make-up plan.

Late Work Policy

Assignments are due by the start of the class period (11:30am) on the due date. For most weeks, this is Thursday (exceptions are noted in the due date calendar, below). Assignments will be penalized by 7% for each day that they are late. Late assignments are accepted up to 7 days late. Adjustments will not be made for non-health, non-academic/extra-curricular reasons (such as mid-semester vacations).

Academic Misconduct

The University of Michigan community functions best when its members treat one another with honesty, fairness, respect, and trust. The college promotes the assumption of personal responsibility and integrity, and prohibits all forms of academic dishonesty and misconduct. In the case of suspected academic misconduct, the student will be given a chance to meet with the instructor and explain their side of the story. If the instructor determines academic misconduct has occurred, the case will be referred to the LSA Office of the Assistant Dean for Undergraduate Education, who will make further determinations. Being found responsible for academic misconduct will usually result in a grade sanction, in addition to any sanction from the college. For more information, including examples of behaviors that are considered academic misconduct and potential sanctions, please see www.lsa.umich.edu/lsa/academics/academic-integrity.html.

Use of Generative AI

ChatGPT, Gemini, Claude, and other similar technologies are advancing rapidly and there are instances where they may be useful in your schoolwork or career. Because of the broad coverage of both textbook and workbook, we ask you to limit your use of generative AI (GenAI) tools. Exceptions to this rule include:

- Personal explorations of the topics (not applied to the course)
- Asking for clarification of the topics or terminology (to contextualize the assignment)

GenAI is not permitted for instances in which students are expected to think critically as part of the learning process. GenAI tools are not permitted in this course for the following activities:

- Answering lab activity or exam questions, including providing short answer responses

- Impersonating you in classroom contexts, such as by using the tool to compose discussion board prompts assigned to you
- Writing entire sentences, paragraphs or papers to complete class assignments.
- Writing emails to the instructors (the professor and GSIs)

If a student is suspected of using GenAI in a manner that is not permitted, the use of GenAI will be treated as an instance of Academic Misconduct (see policy above). If you ever need clarification of what does or does not count as academic misconduct, please ask the instructor.

Grades

Participation/Attendance-Lecture	Participation/Attendance-Lecture	10 percent	10
Participation/Attendance-Lab	Participation/Attendance-Lab	20 percent	20
Homework (Lab) Assignments	10 Lab Assignments (11 Assignments, lowest grade is dropped)	5 percent each	50
Exams	2 Exams	5 percent each	10
	Cumulative Final Exam	10 percent	10
Total Possible Points			100

A+=105-98

A=98-93

A-=92-90

B+=89-87

B=86-83

B-=82-80

C+=79-77

C=76-73

C-=72-70

D+=69-67

D=66-63

D-=62-60

F=59 & lower

Class grades are not curved.

Course Calendar

Week	Day of Week	Date	Lab Due Date	Exam	Reading
1	Tuesday	1-Sep		Canvas Check	Neser Chapter 1, 17
	Thursday	3-Sep	Lab 0: Math Skills Practice (no grade, no stakes)		
2	Tuesday	8-Sep			
	Thursday	10-Sep	Lab 1 Due: Why Study Physical Geography? & Physical Geographer's Tools		
3	Tuesday	15-Sep			Neser Chapter 7-8
	Thursday	17-Sep	Lab 2 Due: Earth History, Geologic Time, Earth's Position in the Solar System		
4	Tuesday	22-Sep			Neser Chapter 14
	Thursday	24-Sep	Lab 3 Due: Atmospheric Science and Meteorology		
5	Tuesday	29-Sep		Exam 1: Lab 1-3	Neser Chapter 2, 4
	Thursday	1-Oct	Lab 4 Due: Internal Processes: Plate Tectonics and Volcanoes		
6	Tuesday	6-Oct			Neser Chapter 9, 13

	Thursday	8-Oct	Lab 5 Due: Internal Processes: Crustal Deformation, Earthquakes, and Landforms		
7	Tuesday	13-Oct			Neser Chapter 5, 10
	Thursday	15-Oct	Lab 6 Due: Weathering and Mass Wasting		
8	Tuesday	20-Oct	<i>Fall Study Break No Classes</i>		
	Thursday	22-Oct			
9	Tuesday	27-Oct			McKnight Chapter 12
	Thursday	29-Oct	Lab 7 Due: Soils		
10	Tuesday	3-Nov		Exam 2: Lab 4-7	McKnight Chapter 11
	Thursday	5-Nov	Lab 8 Due: Climate and Biosphere		
11	Tuesday	10-Nov			Neser Chapter 12
	Thursday	12-Nov	Lab 9 Due: Coastal Processes		
12	Tuesday	17-Nov			Neser Chapter 11
	Thursday	19-Nov	Lab 10 Due: Hydrology		
13	Tuesday	24-Nov	<i>Thanksgiving Break No Classes</i>		
	Thursday	26-Nov			
14	Tuesday	1-Dec			Neser Chapter 15-16
	Thursday	3-Dec	Lab 11 Due: Climate Change		
15	Tuesday	8-Dec			
	Thursday	10-Dec	<i>No Class—Final Exam (Lab 1-11)</i>		

Classroom Culture of Care

LSA is committed to delivering our mission while aiming to protect the health and safety of the community. Our entire LSA community is responsible for protecting the collective health of all members by being mindful and respectful in carrying out the guidelines laid out on the University's Health Response page [<https://healthresponse.umich.edu/>].

In our classrooms all students are expected to adhere to the required safety measures and guidelines of the State of Michigan and the University of Michigan, such as not coming to class when ill or in quarantine. It is important to also be thoughtful about group gatherings as well as about classroom activities and exercises that require collaboration.

Any student who is not able and willing to comply with campus safety measures for this course should contact the course instructor or their academic advisor to discuss alternate participation or course options. For additional information refer to the University of Michigan's Health Response website (link above), and the OSCR Addendum to the Statement of Student Rights and Responsibilities on the OSCR website [<https://oscr.umich.edu/>].

All students, GSI, and the Professor are important and equal partners in this education process regardless of racial/ethnic, cultural, gender, age, social class, caste and other kinds of human difference. We highlight the importance of inclusivity and respect for all students and students' contributions.

Disability Statement

The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact Services for Students with Disabilities [<https://ssd.umich.edu/>; 734-763-3000 or ssdoffice@umich.edu]. For students who are connected with SSD, accommodation requests can be made in Accommodate. If you have any questions or concerns please contact your SSD Coordinator or visit SSD's Current

Student webpage. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience.

Mental Health and Well-Being

University Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressures and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc. If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For personal concerns, U-M offers a variety of resources, many which are listed on the Resources for Student Well-being [<https://wellbeing.studentlife.umich.edu/resources-list>].

Sexual Misconduct Policy

Title IX prohibits discrimination on the basis of sex, which includes sexual misconduct — including harassment, domestic and dating violence, sexual assault, and stalking. We know and understand that sexual violence undermines students' success, and we encourage anyone dealing with sexual misconduct to talk to someone about their experience, so they can get the support they need. Confidential support and academic advocacy can be found with the Sexual Assault Prevention and Awareness Center (SAPAC) on their 24-hour crisis line, 734-936-3333 and at <http://sapac.umich.edu/>.

Alleged violations can be non-confidentially reported to the Equity, Civil Rights, and Title IX Office (ECRT) at ecrtoffice@umich.edu.

Land Acknowledgement

As a class physically hosted at the University of Michigan, we respectfully acknowledge our presence on the traditional and ancestral territory of the Meškwaḡki·aša·hina (Fox), Peoria, Anishinabewaki, and Bodéwadmīkiwen (Potawatomi) peoples. We also acknowledge that lands from the Ojibwe (Chippewa), Odawa (Ottawa), and Bodéwadmīkiwen (Potawatomi) allowed for the creation of the University of Michigan.